

Apr'26 Major Examination: EEL1570 Hardware Software Co-design (OPEN-BOOK)

Guidelines (Total time: 180 minutes, Maximum Marks: 40):

- Please read the question paper very carefully. **NO clarification is required in any question.** In case of any doubt, assume whatever you wish to and state that clearly in your answer.
- Usage of AI tools is NOT allowed.

1. A real-time vision system must process a continuous stream of frames at 75 frames per second (FPS), performing the following operations on each frame: i) Frame acquisition from the camera, ii) Noise filtering (pixel-wise smoothing), iii) Edge detection, iv) Object tagging and classification using a ML model.

You are provided with the following system constraints and resources: The embedded processor can execute 250 million instructions per second (250 MIPS). A custom hardware accelerator (FPGA/ASIC) is available with a budget of 50k logic elements and 24 DSP slices. The total allowable processing time per frame is 13 ms.

Software-only execution time estimates (if all tasks run on CPU):

- Noise filtering: 5 ms per frame
- Edge detection: 7 ms per frame
- Object classification: 10 ms per frame

Hardware implementation estimates are as follows:

- Noise filtering in hardware: 1.5k logic + 5 DSP, latency 0.8 ms
- Edge detection in hardware: 3.5k logic + 7 DSP, latency 1.2 ms
- Object classifier in hardware: 35k logic + 18 DSP, latency 2.5 ms

(a) Propose a hardware–software partitioning strategy such that the 75 FPS real-time constraint is satisfied. Clearly specify which tasks should be mapped to hardware and which should remain in software. Justify your decision using timing and resource considerations.

(b) Based on your proposed partitioning, estimate the total frame processing time and verify whether the system meets the 13 ms per frame constraint. [5 + 3]

2. A deep-tech startup, CipherTech, is developing a proprietary encryption IP for use in multiple security-critical applications. To improve performance and reduce software execution overhead, the company decides to implement part of its encryption algorithm as a dedicated hardware IP block. The algorithm includes the following conditional operation:

```
if (x % 2 == 0) {  
    y = x * x * x;    // cube the number if it is even  
} else {  
    y = x * x;        // square the number if it is odd  
}
```

Design a fully synthesizable and optimized hardware architecture for implementing the above function so that it can be incorporated into encryption IP. Your design should discuss the required datapath, control logic, arithmetic units, input/output bit-width selection, latency, throughput, resource utilization, and possible optimization techniques such as multiplier sharing, pipelining, or parallel computation? [4 + 3]

3. A system-level designer is modeling a simple embedded platform consisting of a CPU, an interconnect, and an external memory. Instead of modeling every clock cycle and signal transition, the designer wants to use Transaction-Level Modeling (TLM) to represent data transfers between the CPU and memory at a higher level of abstraction.

The CPU issues read and write transactions to memory through the interconnect. Each transaction contains an address, command type, data payload, data length, byte-enable information, and response status. The memory responds after a modeled delay, depending on whether the transaction is a read or write and whether the address is valid.

Explain how Transaction-Level Modeling can be used to model this CPU–memory data transfer. In your answer, discuss how a CPU read transaction and a CPU write transaction would be represented, what information should be included in the transaction object, how timing delay can be modeled, and how the memory model should generate responses. Also explain the difference between modeling this system at TLM level and modeling it at RTL level. [3 + 2 + 3]

4. You have been hired by an IITJ-incubated startup to design a speech-command recognition inference engine for a battery-powered wearable device. The inference workload contains feature extraction, matrix-vector operations, activation functions, and decision logic. The available hardware inventory contains:

A baseline CPU with the following characteristics:

- Clock frequency: 1.2 GHz
- Average CPI for integer instructions: 1.2
- Average CPI for floating-point instructions: 3.0
- Dynamic power while executing integer instructions: 0.9 W
- Dynamic power while executing floating-point instructions: 2.1 W
- Static leakage power: 0.35 W
- Area footprint: 0.20 mm²

An ASIC accelerator block that can be placed within the same area budget as the CPU and contains 4 dedicated execution units as described below:

- ALU-FP1: scalar floating-point unit — latency 4 cycles/op, energy 7 nJ/op
- ALU-FP2: vector floating-point unit — latency 2 cycles/op when packed data is available, energy 5 nJ/op
- ALU-INT1: integer ALU — latency 1 cycle/op, energy 1.2 nJ/op
- ALU-INT2: integer ALU — latency 2 cycles/op, energy 1.0 nJ/op
- ASIC block static power: 0.25 W when enabled

The speech-command recognition workload has 10,000 instructions in total. The instruction mix is:

Floating-point instructions: 40%, Integer instructions: 60%

You must decide between the following implementation options: (a) Use the CPU only, (b) Use the ASIC only, where all data-transfer-related instructions are offloaded to the ASIC, (c) Use a hardware/software co-design, where the compiler/runtime maps a fraction of instructions to the ASIC ALUs and the remaining instructions continue to run on the CPU.

For the co-design case, assume: 70% of floating-point instructions are offloaded to the ASIC. 50% of integer instructions are offloaded to the ASIC. The CPU and ASIC operate concurrently. The CPU executes only the non-offloaded instructions while the ASIC executes the offloaded instructions. Ignore communication overhead unless you explicitly state an assumption.

Which implementation option would you choose to achieve maximum performance while minimizing power consumption? Show your calculations for execution time, energy consumption, average power, and any assumptions used. [4+3]

5. A startup is designing an SoC for an edge-based image preprocessing and inference system. The SoC contains an ARM Cortex processor, on-chip SRAM, external DDR memory, a DMA controller, and a custom hardware accelerator for image filtering. The processor configures the accelerator through memory-mapped registers, while image data is transferred between DDR memory and the accelerator using either an AMBA AHB interface or an AXI4 interface.

The system processes grayscale image frames of size 640×480 pixels, where each pixel is 8 bits. For every frame, the accelerator performs filtering and writes the processed image back to DDR memory. Therefore, for each frame:

- One full frame is read from DDR memory.
- One full processed frame is written back to DDR memory.

The following interface options are available:

Option 1: AHB-based design	Option 2: AXI4-based design
AHB data bus width: 32 bits AHB clock frequency: 100 MHz Maximum burst length: 4 beats Bus efficiency due to arbitration and wait states: 65% CPU configuration overhead per frame: 12,000 cycles Accelerator compute time per frame: 1.8 ms AHB interface dynamic power: 120 mW Accelerator dynamic power: 450 mW CPU dynamic power during configuration: 300 mW	AXI data bus width: 64 bits AXI clock frequency: 150 MHz Maximum burst length: 16 beats Bus efficiency due to burst transfer and independent read/write channels: 85% CPU configuration overhead per frame: 18,000 cycles Accelerator compute time per frame: 1.8 ms AXI interface dynamic power: 180 mW Accelerator dynamic power: 450 mW CPU dynamic power during configuration: 300 mW

Answer the following:

- (a) Calculate the total amount of data transferred per frame in bytes. [1]
- (b) For both the AHB-based and AXI4-based designs as mentioned above, calculate the effective memory bandwidth in MB/s. [1]
- (c) Calculate the data-transfer time per frame for both interfaces. [1]
- (d) Calculate the CPU configuration time per frame for both designs.[1]
- (e) Determine whether each design can meet the 60 FPS real-time constraint. [3]
- (f) Estimate the energy consumed per frame for both designs using the given power values. Clearly state any assumptions used for overlapping execution. [3]